Day 134 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Plan and Explain a Statistical Investigation

Standards: 6.SP.A.1, 6.SP.B.4, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d

Learning target: I can plan and Explain a Statistical Investigation, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A statistical question anticipates different answers in a collection of data. Name the group, the measured attribute, and the expected variation. A question about one fixed fact is not statistical. Draw an equally spaced number line and stack one dot per observation above its value. Repeated values stack vertically. Add an axis label with units and check that the dot count equals the data count. The mean is a fair share: add all values and divide by the number of observations. Report the count and measurement units. A large unusual value can pull the mean toward it. A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 13

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01 / Learn

A statistical question anticipates different answers in a collection of data. Name the group, the measured attribute, and the expected variation. A question about one fixed fact is not statistical. Draw an equally spaced number line and stack one dot per observation above its value. Repeated values stack vertically. Add an axis label with units and check that the dot count equals the data count. The mean is a fair share: add all values and divide by the number of observations. Report the count and measurement units. A large unusual value can pull the mean toward it. A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value.

Worked example 1

Is “What are the wingspans, in centimeters, of the birds observed at a feeder?” statistical? Explain the group, attribute, units, and variation.

1. A statistical question anticipates different answers in a collection of data.

2. Name the group, the measured attribute, and the expected variation.

3. The question anticipates a collection of measurements.

4. Each bird can have a different wingspan.

5. Use the same unit for every observation.

6. State the result in the form requested: Yes; observed birds; wingspan; centimeters; wingspans can vary..

Answer: Yes; observed birds; wingspan; centimeters; wingspans can vary.

Worked example 2

Six students recorded reading times (minutes): 3, 6, 6, 9, 9, 12. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Order the observations.

3. Middle positions give median 15/2 and half medians give IQR 3.

4. Mean is 45/6

5. averaging distances gives MAD 5/2.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 15/2, IQR 3; mean 15/2, MAD 5/2. Roughly balanced; mean/MAD or median/IQR with justification.

Day 134 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Is “How many pages did each student read last night?” statistical? Explain.

02 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Draw a labeled dot plot. Check the number of dots.

03 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Report n, units, and mean.

04 / Constructed response

Reading times in minutes are 2, 7, 7, 12, 12, 17. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

05 / Constructed response

Rewrite “How long is my pencil?” as a statistical question.

06 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Draw a labeled dot plot. Check the number of dots.

07 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 7, 16. Report n, units, and mean. If one more student reads 8 minutes, find the new mean.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 11. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Multiple select

A class studies a collection of backpacks. Select all statistical questions, and identify the expected variability.
A. How does mass vary among the backpacks?
B. How many centimeters are in 5 meters?
C. What is the mass in kilograms of each of the backpacks?
D. What is 5 + 3?

10 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Draw a labeled dot plot. Describe a cluster or gap and report the count.

Day 134 | Unit 09 | Answer key

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1. Yes; pages read vary across students.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

2. 7: 1 dot; 8: 2 dots; 9: 2 dots; 10: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

3. n = 6; minutes; mean 15/2 min

Sum = 45; mean = 45 ÷ 6 = 15/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

4. 6 observations of reading time in minutes. Mean 19/2 min, MAD 25/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

5. Example: What are the lengths in centimeters of the pencils in our classroom?

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

6. 3: 1 dot; 8: 2 dots; 13: 2 dots; 18: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

7. n = 6; minutes; mean 43/6 min; new mean 51/7 min

Sum = 43; mean = 43 ÷ 6 = 43/6. New sum = 51; divide by 7.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

8. 6 observations; minutes; median 13/2, IQR 3; mean 13/2, MAD 5/2. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 13/2 and half medians give IQR 3. Mean is 39/6; averaging distances gives MAD 5/2. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

9. A, C - How does mass vary among the backpacks?; What is the mass in kilograms of each of the backpacks?

Different backpacks can have different mass. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

10. 2: 1 dot; 3: 2 dots; 4: 1 dot; 5: 1 dot; 14: 1 dot; n = 6; low cluster and high isolated value

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response